Central Limit Theorem

Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be identical and independent distributed random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. We define

$$Y_n = \sqrt{n}\left(\frac{\bar{X} - \mu}{\sigma}\right) \text{ where } \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Then, the distribution of the function Y_n converges to a standard normal distribution function as $n \to \infty$.

Central Limit Theorem

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Central Limit Theorem Proof

$$X_1, \ldots, X_n \overset{iid}{\sim} F(\theta)$$

$$E(X_i) = \mu < \infty \qquad Var(X_i) = \sigma^2 < \infty$$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

$$\frac{\bar{X}_n - E(\bar{X}_n)}{Var(\bar{X}_n)} \rightarrow \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow N(0,1)$$

$$n \rightarrow \infty$$

WLOG $E(X_i) = \mu = 0$ $Var(X_i) = \sigma^2 = 1$

$$\frac{\bar{X}_n - 0}{1/\sqrt{n}} = \sqrt{n}\,\bar{X}_n = \sqrt{n}\,\frac{1}{n}\sum_{j=1}^{n} X_j$$

$$S_n = \sum_{j=1}^{n} X_j = \frac{\sqrt{n}}{\sqrt{n}}\frac{\sqrt{n}}{n} S_n = \frac{\cancel{n}}{\sqrt{n}\,\cancel{n}} S_n$$

$$\frac{S_n}{\sqrt{n}} \xrightarrow[n \to \infty]{} N(0,1)$$

$$\phi_{S_n/\sqrt{n}}(t) = E\left(e^{i t \frac{S_n}{\sqrt{n}}}\right)$$

$$= E\left(e^{it\frac{1}{\sqrt{n}}\sum_{j=1}^{n} X_j}\right)$$

$$= E\left(\prod_{j=1}^{n} e^{it\frac{1}{\sqrt{n}}X_j}\right) \quad \text{independent}$$

$$= \prod_{j=1}^{n} E\left(e^{it\frac{1}{\sqrt{n}}X_j}\right) \quad \text{identical}$$

$E(X^2) = Var(X) + E(X)^2$

$$\left[E\left(e^{it\frac{1}{\sqrt{n}}X_j}\right)\right]^n \quad \text{Taylor's Series Expansion of } e^x$$

$$\overset{E(1)}{} \quad \overset{it\,E(x)}{} \quad \overset{E(x^2)=1}{}$$

$$\left[E\left(1+\frac{it\,x_j}{1!\sqrt{n}}+\frac{i^2t^2x_j^2}{2n}+\frac{\left(\frac{it\,x_j}{\sqrt{n}}\right)^3}{3!}+\frac{\left(\frac{it\,x_j}{\sqrt{n}}\right)^4}{4!}+\cdots\right)\right]^n$$

$$\left[1+0-\frac{t^2}{2n}-\frac{it^3E(x_j^3)}{6n^{3/2}}+\cdots+\right]^n$$

$$\ln(1+y)=y-\frac{y^2}{2}+\frac{y^3}{3}-\frac{y^4}{4}+\cdots$$

$$y=\frac{i^2t^2}{2n}-\frac{it^3E(x^3)}{6n^{3/2}}+\frac{(it)^4E(x^4)}{4!\,n^{4/2}}+\cdots$$

$$\ln\left[1-\frac{t^2}{2n}-\frac{i t^3 E(x^3)}{6n^{3/2}}+\cdots\right]^n$$

$$n\ln(1+y)=$$

$$n\left[y-\frac{y^2}{2}+\frac{y^3}{3}-\frac{y^4}{4}+\cdots\right]$$

$$n\left[\left(-\frac{t^2}{2n}+\frac{t^3 i^3 E(x^3)}{6n^{3/2}}+\cdots\right)-\left(-\frac{t^2}{2n}+\frac{t^3 i^3 E(x^3)}{6n^{3/2}}+\cdots\right)^2+\cdots\right.$$

Distribute n

$$-\frac{t^2}{2} + \frac{t^3 i^3 E(x^3)}{6n^{1/2}} + \frac{}{n^{3/2}}$$

$$n\left(-\frac{t^2}{2n} + \frac{a}{6n^{3/2}} + \frac{b}{n^{5/2}} + \cdots\right)^2$$

$$n\,\frac{t^2}{2n^2} \qquad -\frac{t^2}{2n}$$

$$n\left(\frac{-t^5}{2n}\right)^3$$

$$n^2$$

$$\left[\left(-\frac{t^2}{2} + \frac{t^3 i^3}{6n^{1/2}} E(x^3) + \cdots\right) + \left(\frac{1}{n} \cdots\right) + \cdots\right.$$

$$\lim_{n\to\infty} -\frac{t^2}{2} + \frac{a}{n^{1/2}} \cdots$$

$$-\frac{t^2}{2} + 0 + 0 \quad 0 \quad 0$$

$$-\frac{t^2}{2}$$

$$\lim_{n\to\infty} \left[\left(-\frac{t^2}{2} + \frac{t^3 i^3}{6n^{3/2}} E(x^3) + \cdots\right)\right] = -\frac{t^2}{2}$$

$$\lim_{n\to\infty} \left[\ln \left[\phi(t) \right] \right] = \frac{-t^2}{2}$$

$$\lim_{n\to\infty} \phi_{\frac{S_n}{\sqrt{n}}}(t) = e^{-\frac{t^2}{2}}$$

$$\phi_X(t) = e^{-t^2/2}$$

$$X \sim N(0,1)$$

Example

Let $X_1, \ldots, X_n \overset{iid}{\sim} \chi^2_p$, the MGF is $M(t) = (1-2t)^{-p/2}$. Find the distribution of $\bar{X}$ as $n \to \infty$.